

Project 2

Metaheuristic Optimisation of TSP Using Genetic Algorithm with Repair Mechanisms

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Problem

Genetic Algorithm

Constraint Handling

Evaluation

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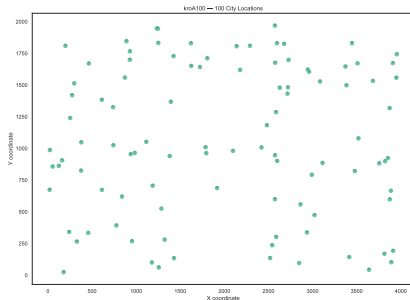
Evaluation

- ▶ 100 cities, symmetric Euclidean distances
- ▶ Known optimal tour length: **21,282**
- ▶ Search space: $\sim 4.7 \times 10^{155}$ distinct tours
- ▶ From TSPLIB (Reinelt, 1991)

Objective:

$$\min_{\pi} f(\pi) = \sum_{i=0}^{n-1} d(\pi_i, \pi_{(i+1) \bmod n})$$

where π is a permutation of $\{0, 1, \dots, 99\}$.



Permutation Encoding

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- ▶ Chromosome = ordered tuple of city indices
- ▶ Example (8 cities): [3, 0, 7, 1, 5, 2, 6, 4]
- ▶ Tour: $3 \rightarrow 0 \rightarrow 7 \rightarrow 1 \rightarrow 5 \rightarrow 2 \rightarrow 6 \rightarrow 4 \rightarrow 3$
- ▶ Fitness = total closed tour length (lower is better)

Permutation Constraint

Every city must appear exactly once. A chromosome with duplicates or missing cities is **not** a valid tour, even though the fitness function still returns a number.

Evolutionary Operators

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Selection

- ▶ Tournament ($k=3$) – primary
- ▶ Roulette wheel – comparison

Crossover

- ▶ Naive single-point – primary
- ▶ PMX – feasibility-preserving comparison

Mutation

- ▶ Swap mutation – sole operator
- ▶ Preserves permutation by construction

Other settings

- ▶ Elitism: top 2 carried over
- ▶ 500 generations, 30 seeds per config
- ▶ Pop size 200, $r_c=0.95$, $r_m=0.10$

Main Loop

```

1: Initialise population  $P$  of  $N$  random permutations
2: Evaluate fitness  $f(p)$  for all  $p \in P$ 
3: for  $g = 1$  to  $G_{\max}$  do
4:    $P' \leftarrow$  top- $e$  elites from  $P$ 
5:   while  $|P'| < N$  do
6:     Select parents  $p_1, p_2$  via tournament or roulette
7:     if  $\text{rand}() < r_c$  then
8:        $c \leftarrow \text{Crossover}(p_1, p_2)$ 
9:     else
10:       $c \leftarrow \text{copy}(p_1)$ 
11:    end if
12:     $c \leftarrow \text{Mutate}(c, r_m)$ 
13:    if repair enabled then
14:       $c \leftarrow \text{Repair}(c)$ 
15:    end if
16:    Append  $c$  to  $P'$ 
17:  end while
18:   $P \leftarrow P'$ ; evaluate fitness for all  $p \in P$ 
19: end for
20: return best individual in  $P$ 

```

▷ elitism

▷ swap

Lines 8 and 14 are the focus of this presentation: what goes wrong at crossover, and how repair fixes it.

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Naive Crossover Breaks the Permutation

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Naive Two-Point Crossover

	0	1	2	3	4	5	6	7
Parent A	0	1	2	3	4	5	6	7
Parent B	2	6	4	0	5	7	1	3

Segment: positions [2, 4]

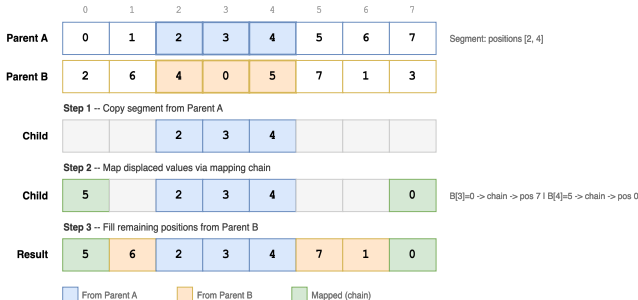
Blind splice -- copy A segment into B (no conflict resolution)

Result	2	6	2	3	4	7	1	3
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Duplicates: {2, 3} -- Missing: {0, 5} -- **INVALID**

 From Parent A
 From Parent B
 Duplicate

Cities {2, 3} duplicated, cities {0, 5} missing. The child is **not** a valid tour – but the fitness function still returns a number, and it tends to be artificially low.

Partially-Mapped Crossover (PMX)

PMX copies the segment from Parent A, then routes displaced values through a mapping chain until each lands in an unoccupied position. Every city appears exactly once – no repair needed.

Our Approach: Repair After the Fact

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Instead of switching to PMX, we keep naive crossover and **repair** every offspring. This makes repair load-bearing on every generation.

Phase 1 – Diagnosis: scan left to right, track *seen* set, flag duplicate positions, compute missing cities.

Phase 2 – Correction (random insertion): shuffle missing cities, place at duplicate positions. No geometric bias, preserves population diversity.

Example

Input: (3, 1, 4, **1**, 5, 2, 6, **3**) Duplicates: {3, 7} Missing: {0, 7}

Random repair → (3, 1, 4, **7**, 5, 2, 6, **0**) ✓

Experiment Setup

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Which strategy handles the permutation constraint best? We run the same GA three ways on kroA100:

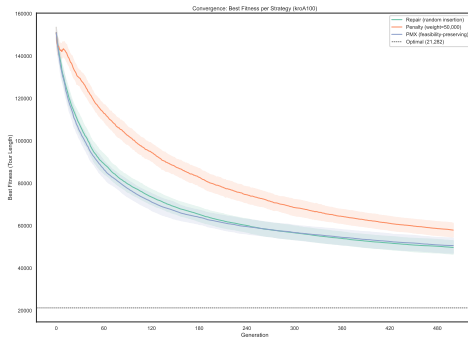
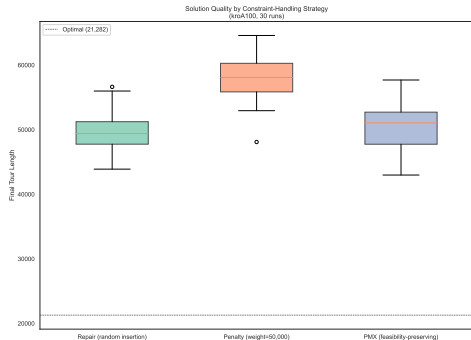
Shared parameters	pop=100, $r_c=0.8$, $r_m=0.2$, tournament, 500 gen
Seeds	30 independent runs per strategy

Repair Naive crossover + random-insertion repair. Every evaluated chromosome is a valid tour.

Penalty Naive crossover, no repair. Infeasible offspring penalised by $w \cdot v$ ($w=50,000$) added to fitness.

PMX Feasibility-preserving crossover. No repair needed.

Results



Strategy	Mean best	Std	Gap
Repair	49,778	3,187	133.9%
PMX	50,655	3,573	138.0%
Penalty	57,948	3,285	172.3%

Repair and PMX are practically tied; penalty trails by $\sim 16\%$.

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1. Naive crossover + repair is competitive with PMX on kroA100 (mean gap differs by $\sim 2\%$)
2. Random insertion preserves diversity better than deterministic repair strategies
3. Penalty function trails both feasibility-based approaches by $\sim 16\%$
4. Enforcing feasibility directly gives the GA a cleaner search signal than penalising infeasibility

Thank you